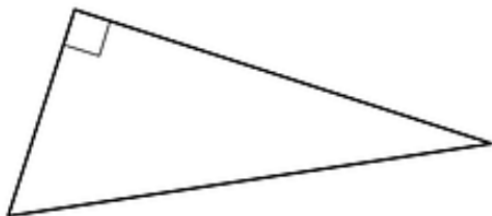


Grade 8  
Unit 5  
Lesson 6 Practice

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

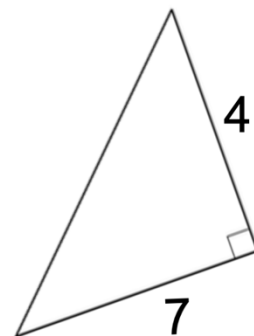
1. For the right triangle below, label the legs  $a$  and  $b$ , and the hypotenuse  $c$ .



Use the right triangle with side length 4 and 7 for questions 2–5.

2. Which side length is unknown?

- ☐ Leg  
☐ Hypotenuse



3. In the equation for the Pythagorean Theorem,  $a^2 + b^2 = c^2$ , substitute the given values for

- ☐  $a$  and  $b$   
☐  $a$  and  $c$   
☐  $b$  and  $c$

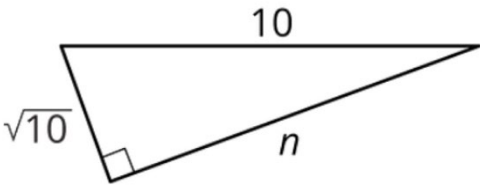
4. Substitute the values in the equation and algebraically solve for the missing variable.

5. Identify the two solutions.

| Exact Solution     | Approximate Solution          |
|--------------------|-------------------------------|
| $c = \sqrt{\quad}$ | $c \approx \underline{\quad}$ |

Use the right triangle with side length 10 and  $\sqrt{10}$  for questions 6–10.

6. Levi wrote the equation  $10^2 + b^2 = \sqrt{10}^2$  to find the missing side length, but made a mistake. What mistake did he make?



7. Which side length is unknown?
- ☐ Leg
  - ☐ Hypotenuse

8. State the Pythagorean Theorem and substitute the given values for
- ☐  $a$  and  $b$
  - ☐  $a$  and  $c$
  - ☐  $b$  and  $c$

9. Solve the equation for the missing variable.

10. Identify the two solutions.

| Exact Solution     | Approximate Solution                 |
|--------------------|--------------------------------------|
| $b = \sqrt{\quad}$ | $b \approx \underline{\hspace{2cm}}$ |